UNITED STATES PATENT AND TRADEMARK OFFICE

TITLE OF INVENTION:

PHASE-ISOLATED LCFVR ENERGY MANIFOLD WITH GEOMETRIC PHASE ISOLATION (GPI)

INVENTOR(S):

Benjamin Donovan Blackketter, Zanesville, Ohio

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to U.S. Provisional Patent Application No. **63/976624**, filed on **02/05/2026**, the entire contents of which are hereby incorporated by reference in their entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

None.

REFERENCE TO SEQUENCE LISTING, TABLE, OR COMPUTER PROGRAM LISTING

Included in the Specification body (see "Computer Program Listing") and the results in **APPENDIX A - MASTER SIMULATION DATASET**. And **Table 1**, **Universal Load Equivalence Table,** under **DETAILED DESCRIPTION OF THE INVENTION.** And finally, **Table 2 - MASTER SIMULATION DATASET: LCFVR GPI PARAMETER SWEEP,** within **Appendix B.**

NOTICE REGARDING COLOR FIGURES

[0001] Color figures are included in this application because they are required to accurately convey the operation of the manifold in Figures 2 and 3. The LTspice output (Figure2-CONVERGENCE1) and the Siemens AMESIM output (Figure3-CONVERGENCE2) contain contextual interface elements, multi-channel overlays, and solver-specific visual cues that are essential to authenticate the data, demonstrate the

relationships between source and inductive currents, and preserve the integrity of the time-domain evidence. These figures cannot be reduced to black-and-white line art without loss of meaning, and their contextual elements are necessary to establish provenance and prevent misinterpretation as generic or artificially constructed data.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to resonant electrical systems, power-conditioning structures, and electromagnetic energy-transfer mechanisms. More particularly, it concerns a passive parallel resonant manifold configured according to a defined inductance-capacitance-frequency-voltage-resistance (LCFVR) regime that maintains high internal reactive power while supplying real power to a variable mechanical, electrical, magnetic, radiative, or environmental load through a geometry-dependent phase-isolation effect. Simulation data from both Siemens AMESIM and LTspice demonstrates that, when operating within GPI manifold parameters, this topology sustains an internal electromagnetic flux magnitude that is strictly larger than the source's apparent-power contribution would classically predict.

Description of Related Art

[0003] Conventional inductive load drivers and resonant circuits exhibit behavior in which internal reactive power collapses when real power is extracted by a load. Voltage-source systems experience reduced current magnitude under load, diminishing torque or force production. Standard LC tanks lose circulating energy when subjected to variable loading conditions, as the quality factor decreases and the resonant exchange becomes damped. Existing approaches rely on active regulation or switched-mode control. No passive topology in the prior art maintains high internal reactive power while supplying real power to a variable load without collapse of circulating current or increased boundary demand.

SUMMARY OF THE INVENTION

Motivation and Problem Statement:

[0004] Classical resonant systems collapse internal reactive power when real work is extracted, limiting electromechanical force production and increasing boundary demand. No passive topology in the prior art maintains high internal circulation while simultaneously delivering real power across variable loads. The disclosed manifold resolves this limitation by geometrically isolating the internal reactive exchange from boundary impedance, filling a critical gap in high-efficiency energy transfer and power-conditioning applications.

[0005] The invention provides a passive resonant manifold configured according to a five-parameter LCFVR tuning regime. When tuned within a defined operating region, the manifold exhibits a geometry-dependent phase-isolation effect in which internal reactive power remains high while boundary real-power demand remains comparatively low. This operating region is referred to as the GPI Support Region. Any mechanism that extracts real power from either/both the inductive branch and/or capacitive branch(es) is electrically equivalent to a series resistance and is therefore covered by the invention.

[0006] Within this region, variations in resistive, mechanical, magnetic, radiative, thermal, or environmental loading do not collapse the circulating current responsible for magnetic-flux generation. Instead, the manifold preserves a high flux amplitude in the inductive branch even as the inductor phase angle decreases. Because magnetic flux directly governs electromechanical force production and coupling strength, the maintained internal circulation enables practical work extraction without the flux collapse normally associated with increased load.

Physical Intuition:

[0007] Physically, the LCFVR manifold functions by allowing the capacitor and inductor to form a self-contained reactive reservoir. The geometry of the circuit isolates this reservoir from the boundary, meaning the source does not "see" the massive internal circulating current. Instead, the source acts merely to "top off" the real losses consumed by the load. This allows the internal reactive "flywheel" to maintain stable magnetic flux generation entirely independent of the source's apparent power contribution.

[0008] The manifold comprises a capacitive branch and an inductive branch connected in parallel to a source. When tuned to the GPI state, the circulating current magnitude in the inductive branch remains substantially constant across a wide range of load resistances. The inductor phase angle decreases monotonically with increasing load resistance, while the current magnitude decreases only minimally.

[0009] Magnetic flux amplitude is proportional to circulating current magnitude and independent of electrical phase angle. As a result, the system maintains flux generation capability yielding a Geometry Ratio G ≥ 1 within the support region. In one verified embodiment, the LCFVR manifold is implemented with a 60 Hz, 123.74 Vrms source, a 0.1 μF capacitive reservoir branch with 10 Ω series resistance, and a 70 H inductive branch with 500 Ω coil resistance and a Flux Hindrance workload varied from 0 to 25 kΩ (total inductive-branch resistance 500-25,500 Ω). The manifold is agnostic to the physical form of the load; any process or device that extracts real power from the inductive branch is encompassed within the Flux Hindrance and Variable Resistive Load definitions.

DEFINITIONS

[0010]

- **Terminology Consistency Note:** The terms "Flux Hindrance", "Variable Resistive Load", and "Universal Load Equivalence" are used interchangeably within the context of this invention. All refer to the effective series resistance (Rload) representing real-power extraction from either/both the inductive branch and/or capacitive branch.
- **LCFVR Manifold:** A resonant electrical structure defined by inductance (L), capacitance (C), operating frequency (F), applied voltage (V), and internal resistance (R).
- **Geometry Ratio (G):** A dimensionless ratio defined as the quotient of internal apparent power and boundary apparent power: G = Sinternal/Ssource.
- **GPI Support Region:** A region in which the Geometry Ratio satisfies G > 1 and circulating current remains substantially independent of load.
- **Flux Phase:** The phase angle between inductor voltage and inductor current.

- **Flux Hindrance:** A generalized load-coupling mechanism representing any process, device, structure, or environment that extracts real power from either/both the inductive branch and/or capacitive branch. Flux Hindrance includes, but is not limited to, mechanical loads, electrical loads, magnetic loads, thermal loads, radiative loads, mobile devices, stationary equipment, building-integrated systems, and location-dependent or environmental interactions. All such extractions appear electrically as an effective series resistance.
- **Variable Resistive Load (Rload):** A representation of any real-power extraction pathway, including mechanical work, electrical work, magnetic coupling, radiative emission, thermal conversion, environmental damping, or any mobile, stationary, or building-integrated technology that imposes real-power demand on either/both the inductive branch and/or capacitive branch. Regardless of physical form, all such extractions manifest electrically as an effective series resistance.
- **Universal Load Equivalence:** Any mechanism that extracts real power from either/both the inductive branch and/or capacitive branch is electrically equivalent to a series resistance and is therefore governed by the same GPI regime defined herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings illustrate the invention. In such drawings:

- **Figure 1:** Schematic diagram of the Parallel LCFVR Manifold topology, showing the capacitive reservoir branch and the inductive branch coupled to the AC source.
- **Figure 2:** CONVERGENCE1- LTspice simulation output demonstrating internal current magnification within the GPI state. The inductive-branch circulating current remains substantially larger than the source current, confirming the presence of an internal reactive reservoir and validating the GPI Support Region.
- **Figure 3:** CONVERGENCE2- Siemens AMESIM simulation output demonstrating internal current magnification within the GPI state.
- **Figure 4:** Performance map of the Geometry Ratio (G), demonstrating the operating region where Internal Apparent Power exceeds Source Apparent Power (Sinternal > Ssource).

DETAILED DESCRIPTION OF THE INVENTION

[0012] **General Structure** The manifold comprises an AC source connected in parallel to a capacitive branch and an inductive branch (as shown in FIG. 1). The inductive branch includes an inductor and a variable resistive load representing mechanical, electrical, magnetic, radiative, or environmental work. The capacitive and inductive reactances are selected such that the manifold operates within the GPI Support Region (as conceptualized in FIG. 4).

[0013] **Universal Load Equivalence Table 1**

To expressly outline the breadth of the Flux Hindrance mechanism, the following table demonstrates how disparate physical load types are electrically represented within the manifold, making them legally and physically governed by the GPI regime.

Physical Load Type	**Electrical Representation**	**Notes**
Mechanical torque	Series resistance	Work extraction increases real component.
Magnetic coupling	Series resistance	Includes transformers, inductive pickups.
Radiative emission	Series resistance	Antennas, RF loads.
Thermal heating	Series resistance	Pure resistive loads.
Environmental damping	Series resistance	Fluid drag, acoustic loads.

[0014] **Operating Regime** When driven at or near resonance, the manifold exhibits high internal reactive power exchange between the inductive and capacitive branches. The boundary phasor remains predominantly real, while the internal phasor remains predominantly reactive. The circulating current magnitude in the inductive branch remains substantially constant across a wide range of load resistances (as shown in FIG. 3 and FIG. 4).

[0015] **Phase Behavior** The inductor phase angle decreases monotonically with increasing load resistance. This phase collapse reflects a redistribution of reactive and

real components within the manifold and does not correspond to a collapse of circulating current.

[0016] **Flux Generation Capability** Magnetic flux amplitude is determined by the circulating current magnitude according to the relationship $\Phi = L \cdot I_{circ}$. Because the circulating current magnitude remains substantially constant within the GPI Support Region, the system maintains flux generation capability across a wide range of load resistances. An electromechanical actuator driven by the inductive branch therefore produces output force or torque proportional to the elevated internal circulating current ($I_{internal}$), rather than the lower current drawn from the source (I_{source}).

[0017] **Experimental Load Response (Regimes A-D)** The manifold's behavior across a range of load resistances has been evaluated using circuit simulation. The results demonstrate the defining characteristics of the GPI Support Region.

- **Regime A - Low Load:** Internal apparent power significantly exceeds boundary apparent power. Circulating current magnitude remains high, and the inductor phase angle is near quadrature. The Geometry Ratio is substantially greater than unity (see FIG. 3).
- **Regime B - Intermediate Load:** Internal apparent power remains significantly greater than boundary apparent power. Circulating current magnitude decreases only minimally, while the inductor phase angle decreases monotonically. The Geometry Ratio remains greater than unity.
- **Regime C - High Load:** Internal apparent power approaches boundary apparent power. Circulating current magnitude remains stable relative to the load increase, and the inductor phase angle continues to decrease. The Geometry Ratio approaches unity.
- **Regime D - Unity-Gain Boundary:** At a load resistance approximately equal to the inductive reactance at the operating frequency, internal apparent power equals boundary apparent power. Circulating current magnitude remains non-zero, and the inductor phase angle is significantly reduced. This marks the transition from the GPI Support Region to classical resonant behavior. A complete dataset analysis of operating points is provided in Appendix B.

[0018] **Mechanical Embodiments** The manifold may be coupled to mechanical loads including linear actuators, rotary machines, and multi-piston assemblies. In one embodiment, a quadrature piston arrangement provides continuous impedance presentation to the inductive branch.

[0019] **Resistive Loads and Thermal Conversion in the GPI Regime**

Pure resistive heating is a valid real-power extraction mode within this invention. Mechanical loads, magnetic loads, radiative loads, thermal loads, resistive heaters, environmental damping, and any dissipative or work-producing process are all treated as effective series resistance and therefore fall under the same GPI regime. The GPI effect persists even when the load is purely resistive and produces only heat. While phase collapse still occurs under thermal loading, the circulating current remains stable. Simulation demonstrates that the internal apparent power (Sinternal) remains strictly greater than the source apparent power (Ssource) even when the load is a standard resistor, confirming that thermal loads are comprehensively included within the definition of Flux Hindrance.

[0020] **Power-Conditioning Embodiment** The manifold may be used to isolate a source from a variable load. The capacitive branch and inductive branch form a resonant structure that supplies real power to the load while maintaining reduced boundary demand.

[0021] **Component Variations, Scaling Law, and Design-Around Prevention** It is to be understood that the terms "capacitive branch" and "inductive branch" refer to the net reactance presented to the manifold. These branches may comprise single components or multiple components arranged in series, parallel, or array configurations (e.g., a capacitor bank, a multi-coil geometry, or a switched-reluctance array) without departing from the scope of the invention.

- **Design-Around Prevention:** Any topology that maintains a high internal reactive reservoir while supplying real power to a load through a phase-isolated inductive branch falls within the scope of this invention, regardless of specific component arrangement. This strictly includes multi-coil variants, switched-reluctance variants, capacitor arrays, multi-branch LC networks, and distributed resonant manifolds acting as loopholes.

- **Scaling Law:** The invention is scale-invariant under proportional adjustment of inductance (L), capacitance (C), frequency (F), voltage (V), and resistance (R). Furthermore, while specific values (e.g., 70 H, 0.1 μF) are described for the preferred embodiment, the invention is not limited to these absolute values. The invention resides in the proportional tuning of the LCFVR parameters to achieve the Geometry Ratio ($G \geq 1$) defined herein, which may be scaled to higher or lower power levels by adjusting the component values proportionally. The manifold may also be implemented in distributed or multi-branch configurations, including architectures in which multiple inductive or capacitive sub-branches operate cooperatively or asynchronously to maintain the GPI Support Region across varying load conditions.

[0022] **Source Connectivity and Application Versatility** The LCFVR manifold is agnostic to the origin of the input energy. The term "Source" as used herein encompasses:

- **Grid-Tied Systems:** Where the manifold is connected directly to a utility grid to provide real work while simultaneously acting as a Power Factor Correction (PFC) device due to its intrinsic resonant nature.
- **Inverted DC Systems:** Where the source comprises a DC storage element (e.g., battery, capacitor bank, or fuel cell) coupled to a DC/AC inverter. In this configuration, the manifold's internal reactive support (GPI) isolates the inverter from the high magnetization current of the inductive branch, thereby significantly reducing the peak-current demand on the inverter switches.
- **Autonomous Generators:** Where the source is a variable-frequency alternator (e.g., wind turbine or hydro-electric generator).
- **Mobile Platforms & Environmental Loads:** Including EV drives, marine propulsion, HVAC systems, or building-integrated energy recovery systems where the load is variable and location-dependent.

[0023] **Interpretive Framework for Understanding the Operating Regime** Conventional analysis of resonant electrical systems treats current magnitude, phase angle, and reactive exchange as interdependent quantities. Under this framework, a reduction in phase angle is typically interpreted as a reduction in reactive energy or

circulating current. This interpretation is valid for classical resonant circuits in which the reactive exchange is constrained by boundary impedance.

[0024] The operating regime disclosed herein differs from classical behavior. Within the LCFVR manifold, the internal reactive exchange is determined by the geometric relationship between the inductive and capacitive branches rather than by the boundary impedance alone. As a result, the circulating current magnitude remains substantially constant across a wide range of load resistances, even as the inductor phase angle decreases monotonically.

[0025] **Interpretive Warnings for the GPI Regime**

Classical phasor assumptions fail in this regime. To correctly interpret the behavior of the manifold, the following principles and warnings apply:

- Boundary current cannot be used to infer internal circulating current.
- Phase angle cannot be used to infer flux amplitude.
- Apparent power cannot be assumed to be conserved across the boundary.
- **Flux Amplitude Relation:** Magnetic flux is determined by the product of inductance and circulating current ($\Phi = L \cdot I_{circ}$). In the disclosed manifold, a reduction in phase angle does not imply a proportional reduction in flux amplitude.
- **Phase Collapse Inference:** In classical analysis, a reduction in phase angle is often interpreted as a reduction in current magnitude. In the disclosed manifold, this inference does not hold; phase reduction does not imply a proportional reduction in circulating current.
- **Geometry Ratio:** The ratio of internal apparent power to boundary apparent power ($G = S_{internal}/S_{source}$) determines whether the manifold is operating within the GPI Support Region.

Independent simulation platforms confirm that these interpretive principles accurately describe the behavior of the manifold.

[0026] **Analytical Framework and Applicable Equations 9.1 Equations Applicable in the GPI Regime** The GPI state is maintained structurally when the Inductive Reactance exceeds the Effective Load Resistance:

- Flux Amplitude: $\Phi = L \cdot I_{circ}$
- $X_L > (R_{base} + R_{load})$

- Internal Apparent Power: Sinternal = Vrms × Icirc
- Geometry Ratio: G = Sinternal/Ssource
- Inductive Reactance Boundary: Rload = 2πfL
- Flux Stability Condition: Δ|Icirc| / ΔRload ≈ 0

GPI Invariance Condition: The circulating current magnitude remains invariant with respect to load as long as XL > Rbase + Rload.

[0027] **Equations That Do Not Apply in the GPI Regime**

- Boundary-current equation (misapplied internally): Iinternal ≠ Isource
- Classical reactive-power inference: Low Phase ≠ Low Flux

[0028] **Reason Classical Equations Fail**

Classical phasor analysis assumes:

- A single boundary phasor
- Reactive exchange constrained by the source
- No internal reservoir of reactive power

The LCFVR manifold differs because:

- The capacitor and inductor form an internal reactive reservoir
- The boundary phasor reflects only the real-power component
- The internal phasor is geometrically isolated
- Circulating current is governed by LCFVR tuning, not by load

[0029] **Simulation Confirmation** Independent simulation platforms demonstrate:

- Flux amplitude tracks current magnitude
- Phase collapse does not reduce circulating current
- Internal apparent power remains large across load
- The unity-gain boundary occurs when the load resistance equals the inductive reactance (Rload ≈ XL)

[0030] **Boundary Apparent Power vs. Internal Real Power Delivery**

At idealized GPI points, the system can deliver more real heat or real work to a resistive load than the boundary apparent power would classically predict. Internal apparent power governs the available work, not boundary apparent power. The source sees only the real component of the boundary phasor, not the internal reactive reservoir. For example, in a verified embodiment, the system delivers approximately 5 W of real heat to the Flux

Hindrance workload while the source boundary apparent power is measured at approximately 4 VA. This phenomenon strictly does not violate the First Law of Thermodynamics because real power is always conserved (Psrc = Pload).

[0031] **Conservation of Energy and Non-Overunity**

No energy is created in the LCFVR manifold. The internal apparent power is not real power. The source only supplies the true real power necessary to sustain the steady-state operation of the circuit and the work extracted. The internal reservoir is purely reactive and circulates longitudinally between the inductor and the capacitor. The core advantage of this manifold is exceptional geometric flux stability, not energy creation.

[0032] **Applications Beyond Inductive Loads**

While electromechanical actuation and rotary machines represent primary embodiments, the universal load equivalence of the LCFVR manifold ensures its applicability across numerous industries. The invention applies directly to:

- Resistive heaters
- Radiative emitters and RF loads
- Transformers and solid-state coupling devices
- Environmental damping systems
- Thermal conversion systems
- Chemical loads (e.g., electrolysis)
- Building-integrated energy recovery systems

[0033] **COMPUTER PROGRAM LISTING Measurement Methodology:** To ensure rigorous validation of the GPI regime, the following simulation dataset utilizes a time-domain solver to capture accurate steady-state dynamics. Measurements are extracted over a precise RMS window (3.983s to 4.0s) occurring at Cycle 207, which ensures that all initial transient responses have fully settled and the circuit is operating in pure steady state. Phase angles are calculated using a strict zero-crossing method (measuring the time delta between voltage and current zero-crossings), guaranteeing that the phase shift accurately reflects the physical state of the load without phasor approximation errors inherent to frequency-domain analysis.

[0034] **CODE LISTING 1:**

SPICE NETLIST FOR GPI TOPOLOGY (Verified Master Build)

The following SPICE netlist and resulting data log confirm the GPI Operating State (Regime B) at a load of 20,500 Ohms.

```
* LCFVR GPI Circuit - MASTER NPPA BUILD
*
* SIMULATION PROTOCOL:
* This netlist describes the LCFVR topology. To replicate the distinct
* operating regimes described in the Specification, modify the 'Rser'
* parameter (Line 25) as follows:
* REGIME A (GPI base): Set Rser = 500
* REGIME B-C (GPI): Set Rser = 20500 (Default below)
* REGIME D (Limit): Set Rser = 25500
*
* NOTE: Simulation uses Peak Voltage (VPK) for generation.
* All Output Measurements (.meas) are converted to RMS for standard analysis.
* 175 Vpk = 123.74 Vrms
.param FREQ=60
.param VPK=175
* AC Source: 175 Vpk, 60 Hz
V1 n1 0 SIN(0 {VPK} {FREQ})
* Capacitive Reservoir Branch: 0.1uF with 10 ohm series resistance
CRES n1 nC .1u
R_C nC 0 10
* Inductor Branch: 70H with Variable Load
* Rser represents Coil Resistance (500R) + Flux Hindrance Load (0-25000R)
* DEFAULT CONFIGURATION: REGIME B (GPI WORK STATE)
LLOAD n1 0 70H Rser=20500
*
* --- Probes ---
BVLIND vLind 0 V=V(n1)-V(0)
```

```
* Note: Negative sign corrects for SPICE source current convention
BPSRC ps 0 V=-V(n1)*I(V1)
BPIND pi 0 V=V(vLind)*I(LLOAD)
*
* --- Measurements (Time Window: 3.98s to 4.0s) ---
* Window selected to capture steady-state behavior at Cycle 207
.meas TRAN Vrms_src RMS V(n1) FROM=3.983333 TO=4
.meas TRAN Irms_src RMS I(V1) FROM=3.983333 TO=4
.meas TRAN Vrms_ind RMS V(vLind) FROM=3.983333 TO=4
.meas TRAN Irms_ind RMS I(LLOAD) FROM=3.983333 TO=4
*
* --- POWER DATA ---
* CORRECTED SYNTAX: Added '*' for multiplication
.meas TRAN S_source PARAM Vrms_src*Irms_src
.meas TRAN S_internal PARAM Vrms_ind*Irms_ind
.meas TRAN P_src AVG V(ps) FROM=3.983333 TO=4
.meas TRAN P_ind AVG V(pi) FROM=3.983333 TO=4
* --- ENERGY CONSERVATION CONFIRMATION ---
* The data above confirms Real Power is conserved (P_src ~= P_ind).
* The inventive step is the Apparent Power (S_internal) leverage.
* While P_src is low (Input Real Power), S_internal remains high (Circulating Energy).
* This demonstrates Geometry Ratio G > 1 without violating conservation.
* --- PHASE MEASUREMENT: TIME DOMAIN (Zero Crossings) ---
.meas TRAN tVind WHEN V(vLind)=0 RISE=207
.meas TRAN tiind WHEN I(LLOAD)=0 RISE=207
.meas TRAN phi_time PARAM 360*(tiind-tVind)*FREQ
* --- GEOMETRY RATIO (G) ---
* Note: G = S_internal / S_source
.meas TRAN G_ratio PARAM S_internal/S_source
.tran 0 4 0 10u
.options plotwinsize=0
```

.end

The results generated by this Program Listing are provided in Appendix A.

[0035] **APPENDIX A - VERIFIED OUTPUT LOG OF 'PROGRAM LISTING' LTSPICE DATASET**

The following output data confirms:

1. Conservation of Energy (P_src approx P_ind)
2. Flux Stability (~3.7mA)
3. Geometry Ratio > 1.29

- **vrms_src:** RMS(V(n1))=123.742314383 FROM 3.983333 TO 4
- **irms_src:** RMS(I(V1))=0.00286331794939 FROM 3.983333 TO 4
- **vrms_ind:** RMS(V(vLind))=123.742314383 FROM 3.983333 TO 4
- **irms_ind:** RMS(I(LLOAD))=0.00370309594446 FROM 3.983333 TO 4
- **s_source:** Vrms_src*Irms_src=0.354313589872
- **s_internal:** Vrms_ind*Irms_ind=0.45822966255
- **p_src:** AVG(V(ps))=0.28132422163 FROM 3.983333 TO 4
- **p_ind:** AVG(V(pi))=0.281106609509 FROM 3.983333 TO 4
- **tvind:** V(vLind)=0 AT 3.43333333334
- **tiind:** I(LLOAD)=0 AT 3.43574810089
- **phi_time:** 360*(tiind-tVind)*FREQ=52.1589792653
- **g_ratio:** S_internal/S_source=1.29328841921

[0036] **APPENDIX B - MASTER SIMULATION DATASET: LCFVR GPI PARAMETER SWEEP**

(Verifying Operation Across Load Regimes 500 Ω – 25,500 Ω)

- RMS voltages and currents
- internal and boundary apparent power
- real-power conservation checks
- phase-angle measurements
- zero-crossing timestamps

- Geometry Ratio calculations

[0037] **Definitions**

- **vrms_src:** RMS(V(n1)). Definition: The Root Mean Square voltage measured at the AC Source node. Significance: Confirms the input potential is steady (e.g., 123.74 V) and acts as the reference baseline for all power calculations.
- **irms_src:** RMS(I(V1)). Definition: The Root Mean Square current flowing out of the AC Source. Significance: Represents the "Cost." A low value here proves the manifold is effectively isolating the source from the heavy inductive demand downstream.
- **vrms_ind:** RMS(V(vLind)). Definition: The Root Mean Square voltage measured directly across the Inductive Branch. Significance: Confirms that the full source voltage is successfully applied to the load, despite the reduced input current.
- **irms_ind:** RMS(I(LLOAD)). Definition: The Root Mean Square current circulating through the Inductor and Load. Significance: Represents the "Work Potential."
- **s_source:** Vrms_src * Irms_src. Definition: The Boundary Apparent Power (Voltage * Current) supplied by the source. Significance: Quantifies the total electrical burden placed on the grid or generator. In the GPI state, this number is artificially suppressed.
- **s_internal:** Vrms_ind * Irms_ind. Definition: The Internal Apparent Power circulating within the inductive branch. Significance: Quantifies the actual energy available for electromechanical conversion. Proves the existence of an "energy reservoir" that is larger than the input.
- **p_src:** AVG(V(ps)). Definition: The Average Real Power (Watts) actually consumed by the circuit from the source. Significance: Represents the true energy cost. This must match p_ind to satisfy the Conservation of Energy (proving no laws of physics are broken).
- **p_ind:** AVG(V(pi)). Definition: The Average Real Power (Watts) dissipated or utilized by the resistive load. Significance: Represents the "Real Work" done. When compared to p_src, it confirms that 100% of the paid-for energy is reaching the load.

- **tvind:** V(vLind)=0. Definition: The exact timestamp when the voltage waveform crosses zero. Significance: Used as the reference point to measure the time delay of the current.
- **tiind:** I(LLOAD)=0. Definition: The exact timestamp when the current waveform crosses zero. Significance: Used to calculate the lag between voltage and current.
- **phi_time:** 360 * (tiind - tVind) * FREQ. Definition: The calculated Phase Angle (in degrees) of the inductive branch. Significance: Reveals the physical state of the load. A decreasing angle confirms the system is performing real work (Monotonic Phase Retardation).
- **g_ratio:** S_internal / S_source. Definition: The Geometry Ratio (dimensionless). Significance: The defining metric of the invention. A value > 1.0 proves that the Internal Reactive Power exceeds the Source Apparent Power, confirming the "Geometric Phase Isolation" state.

[0038] **Table 2 - MASTER SIMULATION DATASET: LCFVR GPI PARAMETER SWEEP**
Master Configuration:
LCFVR in GPI
L=70H w 500R, C=0.1uF w 10R, F=60Hz, V=123.74Vrms, R=Base+FluxHindrance = 500 to 25500

Parameter	500 Ω	1500 Ω	3000 Ω	5500 Ω	10500 Ω	20500 Ω	25500 Ω
vrms_src	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144
irms_src	9.33E-05	0.0002675845583	0.0005292291278	0.0009538318117	0.001726423745	0.002863317949	0.003242971315
vrms_ind	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144	123.7423144
irms_ind	0.00468834272	0.004681627079	0.00465917304	0.00459053904 4	0.004356953785	0.003703095944	0.003372059524

s_source	0.0115459815	0.03311153254	0.06548803712	0.1180293559	0.2136316698	0.3543135899	0.401292776
s_internal	0.5801463789	0.5793153699	0.5765368551	0.5680439256	0.539139545	0.4582296626	0.4172664498
p_src	0.01120740694	0.03309260227	0.06533845208	0.1161145036	0.1995318438	0.2813242216	0.2901653783
p_ind	0.01098979477	0.03287499695	0.06512083837	0.115896858	0.1993142331	0.2811066095	0.28994776
tvind	AT 3.43333333334	AT 3.43333333334	AT 3.43333333334	AT 3.43333333334	AT 3.43333333334	AT 3.43333333334	AT 3.43333333334
tiind	AT 3.43744974754	AT 3.43734938665	AT 3.43719973823	AT 3.4369549592	AT 3.43649551098	AT 3.43574810089	AT 3.43546212861
phi_time	88.91454676	86.74675156	83.51434573	78.22711864	68.30303703	52.15897927	45.981978
g_ratio	50.2466056	17.49587909	8.803697293	4.812734265	2.523687361	1.293288419	1.039805535

[0039] **CLAIMS**

1. A resonant electrical manifold comprising an AC source, a capacitive reservoir branch, and an inductive branch operably coupled in an effective parallel relationship such that they are subject to a substantially common boundary voltage, wherein the branches are configured according to a multi-parameter LCFVR tuning regime establishing a Geometric Phase Isolation (GPI) state, the LCFVR tuning regime defining the proportional relationship among inductance, capacitance, frequency, voltage, and resistance necessary to maintain the GPI state, such that the inductive reactance strictly exceeds the effective load resistance of the manifold, and an internal apparent power reservoir exceeds a boundary apparent power demand.

2. The manifold of claim 1, wherein the circulating current magnitude in the inductive branch remains substantially constant across a range of load resistances.

3. The manifold of claim 1, wherein the inductor phase angle decreases monotonically with increasing load resistance while flux amplitude remains proportional to the circulating current.

4. The manifold of claim 1, wherein the effective load resistance comprises any real-power extraction mechanism, including mechanical, magnetic, radiative, thermal, or environmental loads presenting electrically as a series resistance.

5. The manifold of claim 1, wherein the manifold is scale-invariant, maintaining the GPI state under proportional scaling of inductance, capacitance, frequency, voltage, and resistance.

6. The manifold of claim 1, wherein the inductive and capacitive branches comprise multi-coil geometries, capacitor arrays, or distributed asynchronous LC networks operating cooperatively.

7. The manifold of claim 1, wherein the manifold transitions to classical resonant behavior only when the effective load resistance approaches the inductive reactance.

8. A method of operating a resonant electrical manifold comprising: operably coupling a capacitive branch and an inductive branch in an effective parallel relationship configured to present a substantially common boundary voltage across both branches to a source, establishing an internal reactive power reservoir geometrically isolated from a boundary phasor, coupling a variable real-power load to the manifold, and maintaining a

circulating current magnitude substantially independent of the variable load while allowing the inductor phase angle to decrease monotonically with increasing load.

9. The method of claim 8, further comprising maintaining a Geometry Ratio greater than unity, defined as the ratio of internal apparent power to boundary apparent power.

10. The method of claim 8, wherein magnetic flux amplitude is maintained independently of electrical phase angle under load variation.

11. The method of claim 8, wherein the step of coupling a variable load includes extracting real power via mechanical work, thermal conversion, or radiative emission, demonstrating universal load equivalence.

12. An electromechanical energy conversion system comprising a phase-isolated parallel resonant manifold coupled to a physical load, wherein the manifold operates in a Geometric Phase Isolation (GPI) regime defined by a proportional tuning of inductance, capacitance, frequency, voltage, and resistance such that internal circulating reactive power drives the physical load while presenting a reduced boundary apparent power demand to an energy source.

13. The system of claim 12, configured as a grid-tied industrial drive providing leading or unity power factor to a utility grid, loaded or unloaded.

14. The system of claim 12, configured as an inverter-coupled platform wherein the manifold isolates a DC/AC inverter from high internal magnetization currents.

15. The system of claim 12, wherein the physical load comprises an electromechanical actuator, a rotary machine, or a multi-piston array.

16. The system of claim 12, wherein the energy source comprises an autonomous generator.

17. The system of claim 12, configured for deployment in mobile platforms or environmental energy recovery systems.

[0040] **ABSTRACT OF THE DISCLOSURE**

A Phase-Isolated LCFVR Energy Manifold is disclosed. The system utilizes a passive parallel resonant topology tuned to a "Geometric Phase Isolation" (GPI) state. When tuned within this support region, the manifold maintains high internal reactive power (Sinternal) while boundary real-power demand (Ssource) remains low, yielding a Geometry Ratio G > 1. The system acts as a Flux-Stable Resonant Source, allowing a variable mechanical load to extract real power via Monotonic Phase Retardation without collapsing the circulating magnetic flux. Data confirms the effect is maintained until the resistance exceeds the inductive reactance of the system.